

1. Define the term artificial intelligence.

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions.

2. Elaborate NLP (natural language processing).

Natural Language Processing (NLP) is a subfield of **Artificial Intelligence (AI)** that focuses on the interaction between computers and human languages. It enables machines to understand, interpret, generate, and manipulate natural language (spoken or written) in a meaningful way.

Key Components of NLP

Sentiment Analysis, Machine Translation, Speech Recognition, Text Generation.

Applications of NLP

Chatbots & Virtual Assistants, Spam Filtering

3. Elaborate computer vision feature of AI.

Computer Vision (CV) is a prominent feature of **Artificial Intelligence (AI)** that enables machines to interpret and understand visual information from the world, just like humans. It involves teaching computers to process, analyse, and make sense of digital images, videos, and other visual data.

Key Techniques of Computer Vision

Image Classification, Object Detection, Facial Recognition, Optical Character Recognition (OCR)

Applications of Computer Vision

Healthcare, Autonomous Vehicles, Security Systems

4. Elaborate predictive analysis feature of AI.

Predictive Analysis is a powerful feature of Artificial Intelligence (AI) that uses statistical techniques, machine learning algorithms, and historical data to predict future outcomes or trends. It helps organizations make data-driven decisions by identifying patterns and estimating future events.

Techniques Used in Predictive Analysis

Regression Analysis, Classification, Neural Networks

Applications of Predictive Analysis

Healthcare – Predicting disease outbreaks and patient readmission rates.

Finance – Credit scoring, stock price forecasting, and fraud detection.

Weather Forecasting – Estimating temperature, rain, and natural disasters.

5. Elaborate speech recognition feature of AI.

Speech Recognition is a key feature of **Artificial Intelligence (AI)** that enables machines to convert spoken language into written text or execute commands based on human speech. It allows computers and digital devices to understand, interpret, and respond to verbal instructions.

Types of Speech Recognition Systems

Speaker-Dependent Systems – Trained to recognize the voice of specific users.

Speaker-Independent Systems – Can recognize speech from any user.

Command and Control Systems – Executes commands based on voice input (e.g., Alexa, Siri).

Applications of Speech Recognition

- **Virtual Assistants** – Siri, Alexa, Google Assistant.
- **Voice-Based Search** – Google Voice Search.
- **Customer Service** – Interactive Voice Response (IVR) systems.
- **Security Systems** – Voice authentication for unlocking devices.

6. Elaborate personalization feature of AI.

Personalization is one of the most impactful features of Artificial Intelligence (AI) that customizes products, services, or content based on individual user preferences, behaviour, and data. AI systems use machine learning algorithms and data analytics to deliver tailored experiences to users, enhancing customer satisfaction and engagement.

Techniques Used in AI-Based Personalization

- **Collaborative Filtering** – Suggests products based on what similar users like.
- **Content-Based Filtering** – Recommends items similar to what the user has previously liked.
- **Contextual Personalization** – Customizes content based on user location, device, or time.
- **Predictive Personalization** – Forecasts future preferences based on past behavior.

Applications of Personalization in AI

- **Streaming Platforms** – Netflix, YouTube, and Spotify recommend personalized content.
- **E-commerce** – Amazon and Flipkart suggest products based on user purchase history.
- **Online Advertising** – Google and Facebook deliver targeted ads.
- **Education** – Adaptive learning systems that customize lessons based on student performance.

7. Explain the importance of decision making feature of AI.

Decision-Making is a crucial feature of **Artificial Intelligence (AI)** that enables machines to analyze data, interpret patterns, and make informed decisions without human intervention. It plays a vital role in automating tasks, improving efficiency, and providing accurate insights across various industries.

Types of AI-Based Decision-Making Systems

- **Expert Systems** – Mimics human expert decision-making (e.g., medical diagnosis).
- **Machine Learning Models** – Predicts outcomes using historical data patterns.
- **Reinforcement Learning** – Learns optimal decisions through trial and error.

Applications of Decision-Making in AI

- **Healthcare** – Disease diagnosis and personalized treatment plans.
- **Finance** – Credit risk assessment, fraud detection, and investment advice.
- **Manufacturing** – Predictive maintenance and quality control.
- **Customer Service** – Automated chatbots and sentiment analysis.

8. Explain the importance of automation and robotics feature of AI.

Automation and Robotics are key features of **Artificial Intelligence (AI)** that involve using machines and software to perform tasks without human intervention. These technologies improve productivity, efficiency, and precision across various industries, making processes faster and more cost-effective.

Automation refers to the process of using AI systems to perform repetitive tasks without human involvement. AI-powered automation can learn, adapt, and improve over time by analyzing data and pattern

Robotics involves the design, construction, and operation of robots that perform physical tasks. When combined with AI, robots become more intelligent and capable of making autonomous decisions based on real-time data.

Applications of Automation and Robotics in AI

- **Manufacturing** – Robotic arms for assembling products.
- **Agriculture** – Drones for crop monitoring and automated irrigation.
- **Logistics** – Autonomous delivery robots and warehouse automation.
- **Smart Homes** – Automatic lighting, temperature control, and security systems.

9. Differentiate between machine learning and deep learning.

Aspect	Machine Learning	Deep Learning
Definition	A subset of AI that enables machines to learn from data and make decisions without being explicitly programmed.	A subset of Machine Learning that uses neural networks with multiple layers to process data and make complex decisions.
Data Dependency	Works well with small to medium datasets.	Requires large amounts of data for better performance.
Feature Engineering	Manual feature extraction by experts is required.	Automatically extracts features from raw data.
Algorithms	Uses algorithms like Decision Trees, Random Forest, and Support Vector Machines.	Uses deep neural networks like CNNs, RNNs, and GANs.
Complexity	Suitable for simpler tasks like spam detection or price prediction.	Suitable for complex tasks like image recognition, natural language processing, and autonomous vehicles.
Training Time	Takes less time to train with smaller datasets.	Takes longer to train due to multiple layers and large datasets.
Hardware Dependency	Can run on conventional CPUs.	Requires GPUs or TPUs for faster computations.
Accuracy	Moderate accuracy for complex tasks.	High accuracy for large-scale and complex tasks.

10. Enlist any 2 supervised machine learning algorithms.

1. **Linear Regression**
2. **Decision Tree Classifier**

11. Enlist any 2 unsupervised machine learning algorithms.

1. **K-Means Clustering**
2. **Principal Component Analysis (PCA)**

12. Illustrate the concept of reinforcement machine learning.

Reinforcement Learning (RL) is a type of **Machine Learning (ML)** where an intelligent agent learns how to make decisions by interacting with its environment to maximize a reward. Unlike supervised or unsupervised learning, reinforcement learning focuses on how an agent should take actions to achieve the highest possible reward over time.

The process of Reinforcement Learning involves three main components:

1. **Agent** – The decision-making system that learns from experience.

2. **Environment** – The external system with which the agent interacts.
3. **Actions** – The choices the agent makes.
4. **Reward** – The feedback received after performing an action (positive or negative).

Working Process

1. The **Agent** observes the **Environment**.
2. It performs an **Action** based on a **Policy**.
3. The environment gives a **Reward** based on the action.
4. The agent updates its knowledge and policy.
5. The process repeats until the agent maximizes its reward.

Example Application of Reinforcement Learning

- **Robotics** – Autonomous robots learning to walk or perform tasks.
- **Gaming** – Training AI to play chess or video games like AlphaGo.

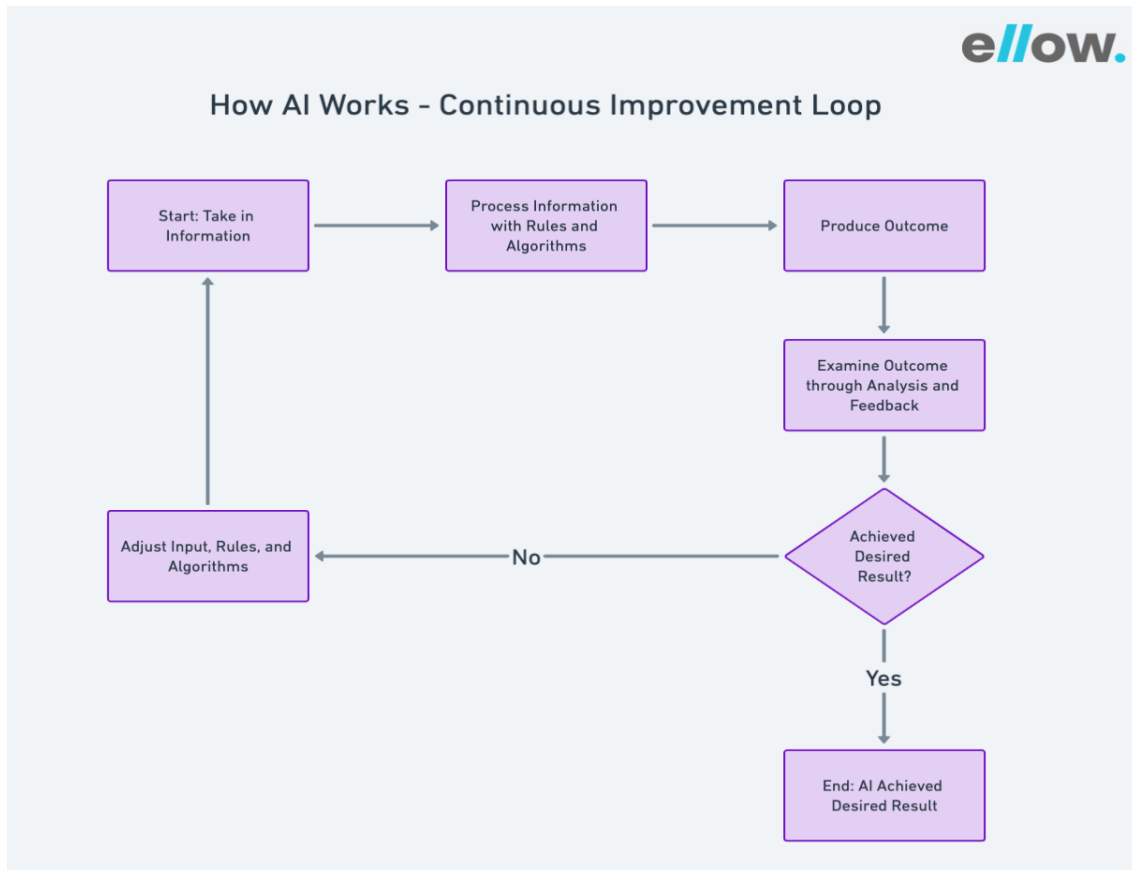
13. Enlist the types of neural networks.

1. Feedforward Neural Network (FNN)
2. Convolutional Neural Network (CNN)
3. Recurrent Neural Network (RNN)
4. Long Short-Term Memory (LSTM)
5. Radial Basis Function Neural Network (RBFNN)
6. Generative Adversarial Network (GAN).
7. Autoencoders
8. Multilayer Perceptron (MLP)

14. Illustrate the working of AI with neat diagram.

AI systems start by taking in information, such as speech, text, or images. Once the data is received, the system uses rules and algorithms to process it. This involves interpreting the information, making predictions, and taking actions based on the input. After processing, the AI system produces an outcome, which could be a success or failure in handling the input data. This result is then carefully examined through analysis, discovery, and feedback to understand how well the AI performed. Using these assessments, the AI system makes adjustments to its input data, rules, and algorithms. This iterative process continues until the AI achieves the desired result. In essence, AI works through a continuous

loop of taking in data, processing it, evaluating the outcome, and refining its approach to improve over time.



15. Define embedded systems.

Embedded Systems are **specialized computer systems** designed to perform specific tasks or functions within larger systems. They consist of both **hardware** and **software** components, working together to perform dedicated functions in real-time.

16. State the applications of embedded systems.

1.consumer Electronics

- Smart TVs
- Digital Cameras

2. Automotive Industry

- Anti-lock Braking System (ABS)
- Airbag Control System

3. Healthcare Industry

- Pacemakers
- MRI Machines

4. Industrial Applications

- Automatic Robotic Arms
- Temperature Control Systems

5. Telecommunications

- Routers
- Modems

6. Aerospace and Defense

- Missile Guidance Systems
- Flight Control Systems

7. Smart Home Applications

- Smart Locks
- Home Security Systems

17. Enlist Key Characteristics of Embedded Systems.(any eight)

1.Dedicated Functionality

- Designed to perform a **specific task** or a limited set of functions.

2. Real-Time Operation

- Capable of processing data and delivering outputs within a defined **time limit**.

3. Reliability

- High reliability to perform critical operations without failure.

4. Compact Size

- Small in size to fit within the host system.

5. Low Power Consumption

- Optimized to consume **minimal power**, especially in battery-operated devices.

6. High Performance

- Provides fast and efficient performance for specific tasks.

7. Embedded Software

- Software (firmware) is permanently stored in **ROM, EEPROM, or Flash memory**.

8. Cost-Effective

- Designed to be **affordable** and mass-produced for commercial use.

9. Real-Time Constraints

- Must perform tasks within strict deadlines, especially in safety-critical applications.

10. Limited User Interface

- Simple or no user interface, often with buttons, LEDs, or small screens.

11. Memory Constraints

- Limited **RAM and storage capacity** due to small size and cost constraints.

12. Microcontroller or Microprocessor-Based

- Powered by a **microcontroller** or **microprocessor** as the central processing unit.

18. Enlist Different architectural components of Embedded Systems.

Hardware Components

- **Processor (CPU)**
 - Microprocessor or Microcontroller that performs all computations and decision-making tasks.
 - Example: ARM, Intel 8051
- **Memory Units**
 - Stores data and instructions needed for the system's operation.
 - Types:
 - ROM (Read Only Memory)
 - RAM (Random Access Memory)
 - Flash Memory
- **Input Devices**
 - Used to collect data from the external environment.
 - Example: Sensors, Keypads, Touch Screens

- **Output Devices**
 - Used to display or communicate results.
 - Example: LEDs, LCDs, Buzzers, Motors
- **Timers & Counters**
 - Used for time delay generation and event counting.
- **Communication Interfaces**
 - Used to transfer data between devices.
 - Example: UART, I2C, SPI, USB
- **Power Supply**
 - Provides the required power to the entire embedded system.

2. Software Components

- **Operating System**
 - Controls hardware components and manages system resources.
 - Example: RTOS (Real-Time Operating System), Linux
- **Firmware**
 - Permanent software programmed into ROM or Flash Memory.
- **Device Drivers**
 - Interface between hardware components and software applications.
- **Application Software**
 - Software designed to perform the desired task or functionality.
- **Middleware**
 - Acts as a bridge between the OS and application software.

19. Write a short note on PIC (Peripheral Interface Controller).

- **Developed by:** Microchip Technology
- **Type:** Microcontroller (MCU)
- **Architecture:** Harvard Architecture, RISC-based
- **Bit-width:** 8-bit, 16-bit, and 32-bit versions
- **Features:**
 - Low power consumption
 - Inbuilt ADC (Analog-to-Digital Converter)
 - Supports serial communication protocols (UART, SPI, I2C)
 - Flash memory for program storage
- **Applications:** Home automation, automotive, industrial control, robotics

Example: PIC16F877A – commonly used in embedded applications.

20. Enlist the features of ARM (Advanced RISC Machine).

- **Developed by:** ARM Holdings
- **Type:** Microprocessor & Microcontroller
- **Architecture:** RISC (Reduced Instruction Set Computing)
- **Bit-width:** 32-bit and 64-bit
- **Features:**
 - High performance with low power consumption
 - Supports multi-core processing
 - Used in embedded Linux, RTOS, and AI applications
 - Commonly found in SoCs (System-on-Chip)
- **Applications:** Smartphones, IoT devices, automotive systems, industrial automation

Example: ARM Cortex-M (MCUs for IoT), ARM Cortex-A (used in mobile processors like Snapdragon).

21. Write a short note on AVR (Alf-Egil Bogen and Vegard Wollan RISC) Microcontroller.

- **Developed by:** Atmel (now part of Microchip Technology)
- **Type:** Microcontroller (MCU)
- **Architecture:** Harvard Architecture, RISC-based
- **Bit-width:** 8-bit and 32-bit
- **Features:**
 - Fast instruction execution (single-cycle execution)
 - Low power consumption (suitable for battery-powered devices)
 - Used in Arduino boards
- **Applications:** Embedded control systems, Arduino-based projects, small IoT devices

Example: ATmega328P (used in Arduino Uno).

22. Write a short note on ASIC (Application-Specific Integrated Circuit).

- **Type:** Custom-designed processor
- **Architecture:** Varies based on application
- **Features:**
 - Optimized for a specific application
 - High performance and low power consumption
 - Cannot be reprogrammed after manufacturing
- **Applications:** Cryptocurrency mining (e.g., Bitcoin miners), networking devices, medical equipment, automotive ECUs

Example: Google's Tensor Processing Unit (TPU) – an ASIC designed for AI applications.

Find Remaining questions answers in notes and ppt shared.

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